

Exercise 8

Team Collaboration Using GitHub — Beginner Lab Manual

What's inside

Step 1	Form a team of 2-3 students and set up Git
Step 2	Create a shared GitHub repository
Step 3	Assign tasks using Issues
Step 4	Work in separate branches
Step 5	Pull Requests and Code Reviews
Step 6	Document the project with README.md

AIM: To collaborate as a team using GitHub: create a shared repository, assign tasks, work in branches, use Issues, Pull Requests and Code Reviews, and document the code with README.md.

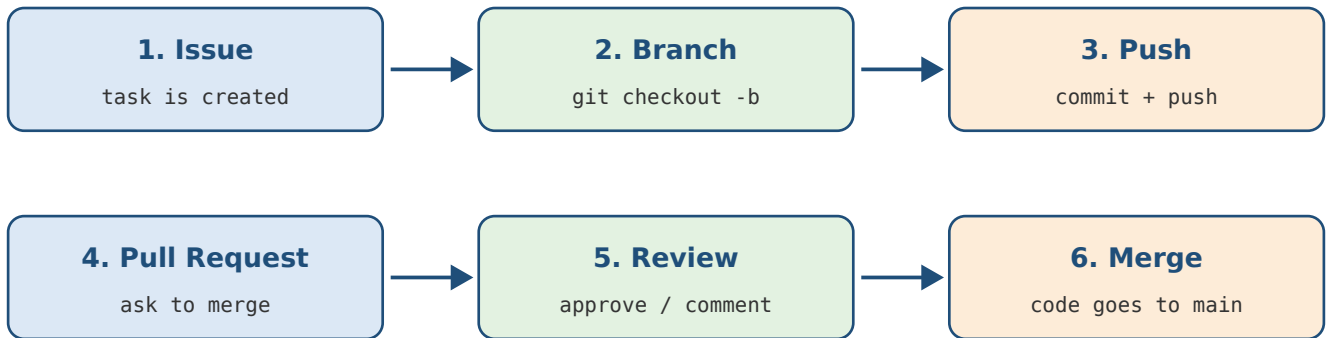
Key words

Word	Meaning (simple)
Repository (repo)	A project folder stored on GitHub
Branch	Your own copy of the code to work on safely
main	The main branch - the final, working code
Commit	A saved snapshot of your changes
Push / Pull	Upload to GitHub / Download from GitHub
Issue	A task or bug written on GitHub
Pull Request (PR)	A request to add your branch's code into main
Code Review	A teammate checks your code before it is merged

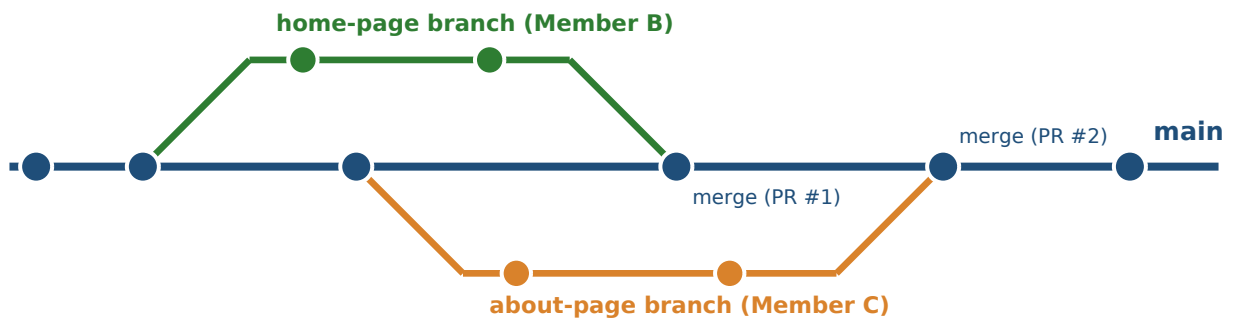
The Big Picture

How a team works together on GitHub

Workflow for every task



Branches in our example



Team roles (sample project: college-website)

Member	Role	Task (Issue)	Branch
Member A	Team Lead	Create repo, write README, review PRs	main
Member B	Developer	#1 Create home page (index.html)	home - page
Member C	Developer	#2 Create about page (about.html)	about - page

NOTE: For a team of 2, Member A also does one of the pages.

Step 1: Set up Git (everyone)

1. Install **Git** from git-scm.com and create a free account on **github.com**.
2. Open a terminal (Git Bash / VS Code) and tell Git who you are:

TERMINAL

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

Step 2: Create a shared repository (Team Lead)

1. On GitHub click + > **New repository**.
2. Name: college-website, choose **Public**, tick **Add a README file**, click **Create repository**.
3. Go to **Settings > Collaborators > Add people** and add your teammates' GitHub usernames.
4. Teammates: accept the invite from your **email** or GitHub notifications.

Everyone: download (clone) the repo

TERMINAL

```
git clone https://github.com/LEAD-USERNAME/college-website.git
cd college-website
```

Step 3: Assign tasks using Issues (Team Lead)

1. Open the repo and click the **Issues** tab > **New issue**.
2. Write a title, e.g. Create home page, and a short description.
3. On the right side, under **Assignees**, choose the teammate. Click **Submit**.

Issue	Title	Assigned to
#1	Create home page	Member B
#2	Create about page	Member C
#3	Write README.md	Member A

Step 4: Work in a branch (each member)

Example shown for Member B. Member C does the same with 'about-page'.

1. Create your branch and switch to it

TERMINAL

```
git checkout -b home-page
```

2. Write your code (index.html)

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Our College</title>
</head>
<body>
  <h1>Welcome to Our College</h1>
  <p>B.Tech CSE Department</p>
  <a href="about.html">About Us</a>
</body>
</html>
```

3. Save (commit) and upload (push)

TERMINAL

```
git add .           # select all changed files
git commit -m "Add home page" # save a snapshot with a message
git push origin home-page # upload your branch to GitHub
```

OUTPUT

```
To https://github.com/LEAD-USERNAME/college-website.git
* [new branch]      home-page -> home-page
```

GOLDEN RULE: Never work directly on **main**. Always make a branch first. That way, nobody breaks the working code.

Step 5: Pull Request and Code Review

Open a Pull Request (Member B)

1. On GitHub, click the yellow **Compare & pull request** button.
2. Check: **base: main** <- **compare: home-page**.
3. Title: Add home page. In the description write Closes #1 (this links the Issue).
4. On the right, under **Reviewers**, choose the Team Lead. Click **Create pull request**.

Review the code (Team Lead)

1. Open the Pull Request and click the **Files changed** tab.
2. To comment on a line, hover over it and click the blue **+**. Example: Please add a page title.
3. Click **Review changes** and choose **Approve** or **Request changes**.

If changes are requested (Member B)

Fix the code on the same branch, then commit and push again. The Pull Request updates automatically.

TERMINAL

```
git add .
git commit -m "Fix: add page title"
git push origin home-page
```

Merge (Team Lead)

1. Click **Merge pull request > Confirm merge**.
2. Click **Delete branch** (the work is now safely in main).
3. Issue #1 closes automatically because of Closes #1.

Everyone: get the latest code

TERMINAL

```
git checkout main
git pull origin main
```

Step 6: Document with README.md (Team Lead)

README.md is the first thing people see on your repo page

Do this the same way: make a branch readme, edit README.md, push, open a PR, and let a teammate review it.

README.md

README.md

College Website

A simple college website made by our team to learn GitHub collaboration.

Team Members

Name	Role	Work Done
Member A	Team Lead	README, reviews
Member B	Developer	Home page
Member C	Developer	About page

Pages

- index.html - Home page
- about.html - About page

How to Run

1. Clone the repo:
git clone https://github.com/LEAD-USERNAME/college-website.git
2. Open index.html in any web browser.

Markdown basics

You type	You get
# Title	Big heading
## Section	Smaller heading
- item	Bullet point
bold	bold text
a b	Table row

Command Cheat Sheet & Final Checklist

Command	What it does
<code>git clone URL</code>	Download a repo to your computer
<code>git checkout -b name</code>	Create a new branch and switch to it
<code>git checkout main</code>	Switch back to main
<code>git status</code>	See which files changed
<code>git add .</code>	Select all changed files
<code>git commit -m "msg"</code>	Save a snapshot with a message
<code>git push origin name</code>	Upload your branch to GitHub
<code>git pull origin main</code>	Download the latest main

Common problems

Problem	Fix
Push rejected	Run <code>git pull origin main</code> first, then push again
Permission denied (403)	Accept the collaborator invite, or ask the lead to add you
Asked for password	GitHub doesn't accept passwords; sign in through the browser window that opens

Final checklist (show this in your lab record)

- ☐ Shared repo created with all team members as collaborators
- ☐ At least 3 Issues created and assigned
- ☐ Each member worked in their own branch
- ☐ Pull Requests opened, reviewed (with comments) and merged
- ☐ All Issues closed
- ☐ README.md written with team details

Result: The team successfully collaborated on a shared GitHub repository using branches, Issues, Pull Requests and Code Reviews, and documented the project with README.md.